

DEPT. EHS	Reactor Startup Procedure	No. Rev No. Date	SOP-REAC-OPS 01 02.24.2026
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1.0	Initial Core Evacuation	Notes
I	The goal of the Initial Core Evacuation is to get the core pressure to below 1×10^{-3} Pa to ready the cryopumps for operation.	
II	For the rest of this manual, the following terms will be used: VALVES • CLOSE - to SHUT OFF a valve • OPEN - to OPEN a valve PUMPS • CLOSE - to TURN ON a pump • TRIP- to TURN OFF a pump CONTROLLERS • INCREASE - to RAISE setpoint • DECREASE - to LOWER setpoint	
1.	CLOSE Roughing Vacuum Pumps (2H) ☐ RV-P-1A ☐ RV-P-2A ☐ RV-P-1B ☐ RV-P-2B	(2H) indicates the panel the equipment is on.
2.	OPEN Roughing Vacuum Valves (2H) ☐ RV-V-1A ☐ RV-V-2A ☐ RV-V-1B ☐ RV-V-2B	This allows the vacuums to discharge to the Core Exhaust Processing system.
	AWAIT 5.10×10^{-3} Pa Core Pressure THEN CONTINUE to section 2.0	

2.0	Cryopump Operations	Notes
I	This section will guide you on how to start up cryopumps to pull a vacuum up to 9.10×10^{-6} Pa.	
1.	TRIP Roughing Vacuum Pumps (2H) ☐ RV-P-1A ☐ RV-P-2A ☐ RV-P-1B ☐ RV-P-2B	Roughing vacuums become ineffective after 5.00×10^{-3} Pa and running them below that will damage them.

2.0	Cryopump Operations (Continued)	Notes
2.	CLOSE Roughing Vacuum Valves (2H) ☐ RV-V-1A ☐ RV-V-2A ☐ RV-V-1B ☐ RV-V-2B	Closing the valves ensures a leak after the valves will not allow air to leak into the reactor.
3.	OPEN Cryopump TIP Suction/Discharge Valves (3H) ☐ CRYO-V-1A ☐ CRYO-V-1B ☐ CRYO-V-2A ☐ CRYO-V-2B	
4.	CLOSE Cryopump Coolant Heat Exchange Pumps (3H) ☐ CRYO-P-3A ☐ CRYO-P-3B ☐ CRYO-P-1A ☐ CRYO-P-1B ☐ CRYO-P-2A ☐ CRYO-P-2B	
5.	DECREASE Cryopump Chiller Controller Setpoint (3H) TO -240 DEG C ☐ CRYO-C-1A ☐ CRYO-C-1B	
	AWAIT 9.10×10^{-6} Pa Core Pressure THEN CONTINUE to section 3.0	

3.0	Core Bake	Notes
I	The purpose of the Core Bake is to vaporize matter off the walls of the core, and to remove the vapor via the Cryopumps.	
1.0	CLOSE Poloidal Field Magnets (4H) ☐ MG-PL-1 ☐ MG-PL-2	
2.0	INCREASE Magnet Current Controller Setpoint (4H) TO 100% ☐ MG-C-1	Both Toroidal and Poloidal field magnets use one controller.
	AWAIT >370 DEG C Core Temperature THEN AWAIT < 100 mg Core Matter THEN CONTINUE to section 4.0	

4.0	Forming a Plasma	Notes
I	In this section, you will form a plasma by injecting fuel matter using the Neutral Beam Injector, which at the same time, will heat up the matter and raising Core Temperatures	
1.0	CLOSE Toroidal Field Magnets (4H) ☐ MG-TD-1 ☐ MG-TD-2 ☐ MG-TD-3 ☐ MG-TD-4 ☐ MG-TD-5 ☐ MG-TD-6 ☐ MG-TD-7 ☐ MG-TD-8	The Magnet Current Controller will run back to 30% due to the Startup Power < 1% Interlock. Annunciators are expected.
2.0	CLOSE Neutral Beam Pumps (5H) ☐ NB-P-1 ☐ NB-P-2	
3.0	INCREASE Injection Controllers (5H) TO 10% ☐ NB-I-1 ☐ NB-I-2	
	AWAIT NB-P-1 and NB-P-2 to stop blinking on Mnemonic Panel (5H) THEN CONTINUE to step 4.0	This will ensure pumps are at the same RPM.
4.0	OPEN Fijium and Manganese Tank Valves (5H) ☐ NB-V-1A ☐ NB-V-2B	Ensure no simultaneous operation of valve sets to prevent fuel damage.
	AWAIT 1.50×10^4 DEG C Core Temperature THEN CONTINUE to step 5.0	A plasma will form at 1.16×10^4
5.0	CLOSE Fijium and Manganese Tank Valves (5H) ☐ NB-V-1A ☐ NB-V-2B	This will stop fuel injection with Neutral Beams.
6.0	TRIP Neutral Beam Pumps (5H) ☐ NB-P-1 ☐ NB-P-2	Trip the pumps so they don't get damaged.
7.0	DECREASE Injection Controllers (5H) TO 0% ☐ NB-I-1 ☐ NB-I-2	Close the injection valves so the fuel doesn't backflow.

5.0	Nuclear Fusion	Notes
I	This step of the process will raise temperatures to 1.6×10^7 where fusion can occur and raise power to 100%.	
1.0	DECREASE Chiller Controllers (6H) TO -240 DEG C ☐ FP-C-1 ☐ FP-C-2	
2.0	INCREASE Injection Controller (6H) TO 1.5% ☐ FP-C-3 ☐ FP-C-4	
3.0	CLOSE Frozen Pellet Pumps (6H) ☐ FP-P-1 ☐ FP-P-2	
4.0	INCREASE Radiofrequency Heating Controller (5H) TO 100A ☐ RF-S-1 ☐ RF-S-2	
	AWAIT Startup Power 1% THEN CONTINUE to step 5.0	
5.0	DECREASE Radiofrequency Heating Controller (5H) TO 0% ☐ FP-C-3 ☐ FP-C-4	
6.0	INCREASE Magnet Current Controller Setpoint (3H) TO 100% ☐ MG-C-1	
	INCREASE Injection Controller (6H) TO 35% ☐ FP-C-3 ☐ FP-C-4	Increase both simultaneously

You've now achieved fusion, to control the power,
INCREASE or **DECREASE** Injection Controller (6H)
And power should move accordingly